Reg. No. | | | | | | | | | | [12CS-IS65/13CS-IS65]

SIXTH SEMESTER B.E/B.TECH EXAMINATION, JUNE-2016

JAVA AND J2EE

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions.

1.	**a.**	Explain how Java is robust and architecture neutral.	**04**
	b.	How arrays are used in Java? Give examples.	**04**
	c.	Explain the following;	**06**
		i. JDK	
		ii. JVM	
		iii. byte code	
2.	**a.**	Discuss the following terms (keywords)	**06**
		i. super	
		ii. this	
		iii. final	
	b.	Write a program in Java using Inheritance to compute the area and volume of an object. Use super keyword in the program. Area = length × breadth Volume = length × breadth × height	**08**
3.	**a.**	Describe the methodology for creation and using a package within Java program.	**06**
	b.	Create a try block that is likely to generate three types of exceptions and then incorporate necessary catch block to handle them.	**08**
4.	**a.**	How Synchronization can be achieved for threads in java? Explain with syntax?	**04**
	b.	Explain the mechanism of delegation event model. List the use of adapter class with an example.	**10**
5.	**a.**	What is an applet? Explain the different stages in the life cycle of an applet.	**06**
	b.	Provide a brief description with syntax about 2 AWT components and 2 layout managers.	**08**
6.	**a.**	Describe the two key features of swing.	**04**
	b.	Write a program using swing to create four different types of buttons. Use suitable events to show actions on buttons and use label to display the action involved.	**10**

7. a. What are database drivers? Briefly discuss about the different types of drivers used in JDBC. **08**

b. Give Java Syntax for the following **06**

i. Connectivity to a database using JDBC/ODBC bridge which has url="Jabc:OdBc:SDB", username = "Scott" and password= "tiger"

ii. Running a query "select emp-age from emp" over the connected database.

iii. Counting number of employees, with condition emp-age grater then 50.

8. a. Write a simple basic Java Servlet program to print "Hello" and describe the steps involved in the execution of this Servlet. **08**

b. Distinguish between CGI and Servlets by giving three differences. **06**